

Orthogonal Functions and Fourier Series



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Vector Analogy

❖ Recalling from the vectors in 3-space that

$$\mathbf{u} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3, \quad (4)$$

we have

$$\mathbf{u} = \frac{(\mathbf{u}, \mathbf{v}_1)}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 + \frac{(\mathbf{u}, \mathbf{v}_2)}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 + \frac{(\mathbf{u}, \mathbf{v}_3)}{\|\mathbf{v}_3\|^2} \mathbf{v}_3 = \sum_{n=1}^3 \frac{(\mathbf{u}, \mathbf{v}_n)}{\|\mathbf{v}_n\|^2} \mathbf{v}_n \quad (5)$$

Thus we can make an analogy between vectors and functions.



12.1 Orthogonal Functions

DEFINITION 12.1

Inner Product of Function

The *inner product* of two functions f_1 and f_2 on an interval $[a, b]$ is the number

$$(f_1, f_2) = \int_a^b f_1(x) f_2(x) dx$$

DEFINITION 12.2

Orthogonal Function

Two functions f_1 and f_2 are said to be *orthogonal* on an interval $[a, b]$ if

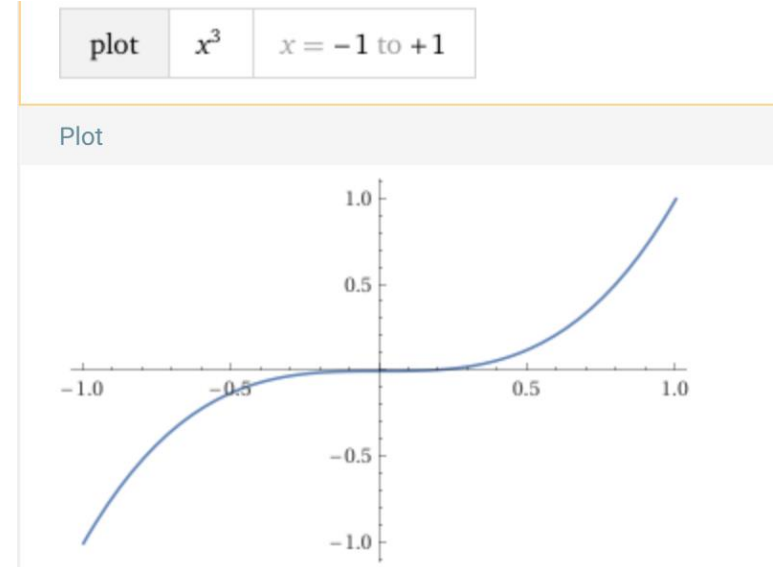
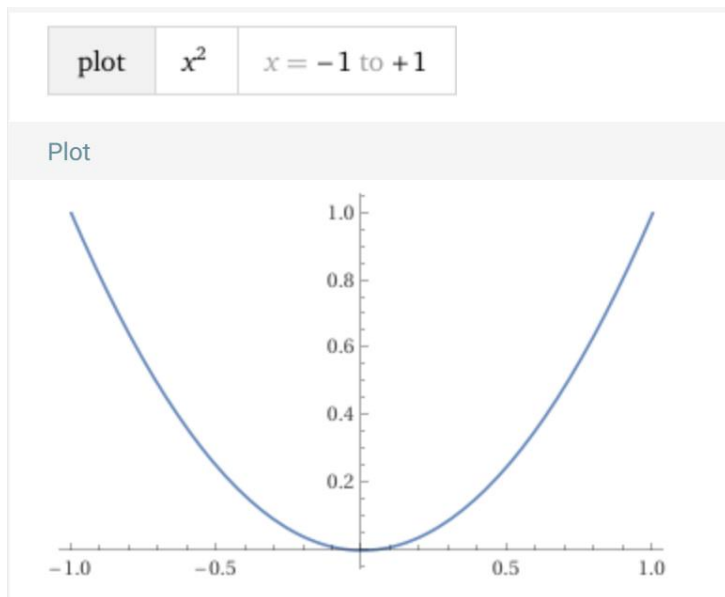
$$(f_1, f_2) = \int_a^b f_1(x) f_2(x) dx = 0$$



Example

❖ The function $f_1(x) = x^2$, $f_2(x) = x^3$ are orthogonal on the interval $[-1, 1]$ since

$$(f_1, f_2) = \int_{-1}^1 x^2 \cdot x^3 dx = \frac{1}{6} x^6 \Big|_{-1}^1 = 0$$





DEFINITION 12.3

Inner Product of Function

A set of real-valued functions $\{\phi_0(x), \phi_1(x), \phi_2(x), \dots\}$ is said to be **orthogonal** on an interval $[a, b]$ if

$$(\phi_m, \phi_n) = \int_a^b \phi_m(x), \phi_n(x) dx = 0, \quad m \neq n \quad (2)$$



Orthonormal Sets

- ❖ The expression $(\mathbf{u}, \mathbf{u}) = \|\mathbf{u}\|^2$ is called the ***square norm***. Thus we can define the square norm of a function as

$$\|\phi_n(x)\|^2 = \int_a^b \phi_n^2 dx, \quad \|\phi_n(x)\| = \sqrt{\int_a^b \phi_n^2(x) dx} \quad (3)$$

If $\{\phi_n(x)\}$ is an orthogonal set on $[a, b]$ with the property that $\|\phi_n(x)\| = 1$ for all n , then it is called an ***orthonormal set*** on $[a, b]$.



Example 1

Show that the set $\{1, \cos x, \cos 2x, \dots\}$ is orthogonal on $[-\pi, \pi]$.

Solution

Let $\phi_0(x) = 1$, $\phi_n(x) = \cos nx$, we show that

$$\begin{aligned}(\phi_0, \phi_n) &= \int_{-\pi}^{\pi} \phi_0(x) \phi_n(x) dx = \int_{-\pi}^{\pi} \cos nx dx \\ &= \frac{1}{n} \sin nx \Big|_{-\pi}^{\pi} = 0, \text{ for } n \neq 0\end{aligned}$$



Example 1 (2)

and

$$\begin{aligned}(\phi_m, \phi_n) &= \int_{-\pi}^{\pi} \phi_m(x) \phi_n(x) dx = \int_{-\pi}^{\pi} \cos mx \cdot \cos nx dx \\&= \frac{1}{2} \int_{-\pi}^{\pi} [\cos(m+n)x + \cos(m-n)x] dx \\&= \frac{1}{2} \left[\frac{\sin(m+n)x}{m+n} + \frac{\sin(m-n)x}{m-n} \right] \Bigg|_{-\pi}^{\pi} = 0, m \neq n\end{aligned}$$



Example 2

Find the norms of each functions in Example 1.

Solution

$$\phi_0 = 1, \|\phi_0\|^2 = \int_{-\pi}^{\pi} dx = 2\pi \Rightarrow \|\phi_0\| = \sqrt{2\pi}$$

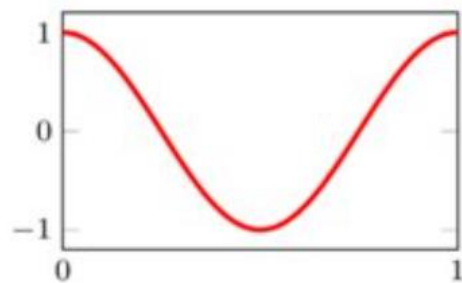
$$\phi_n = \cos nx,$$

$$\|\phi_n\|^2 = \int_{-\pi}^{\pi} \cos^2 nx dx = \frac{1}{2} \int_{-\pi}^{\pi} (1 + \cos 2nx) dx = \pi$$

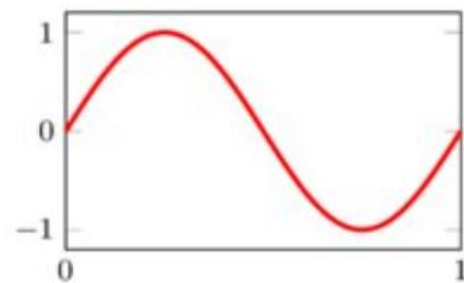
$$\|\phi_n\| = \sqrt{\pi}, n > 0$$



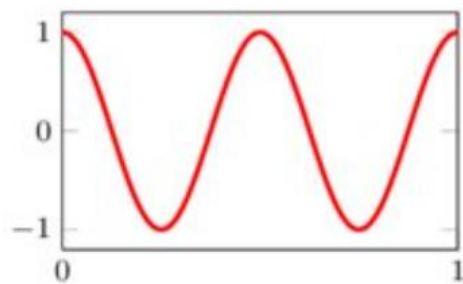
(a) constant term



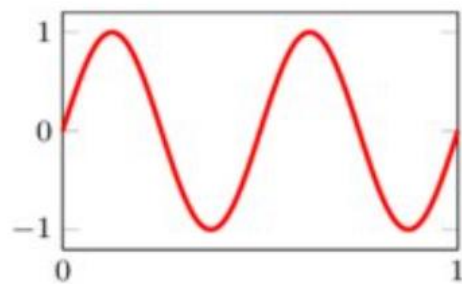
(b) $\cos(2\pi t)$



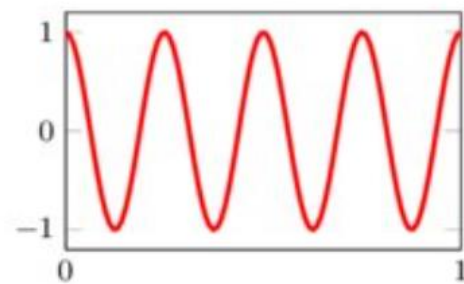
(c) $\sin(2\pi t)$



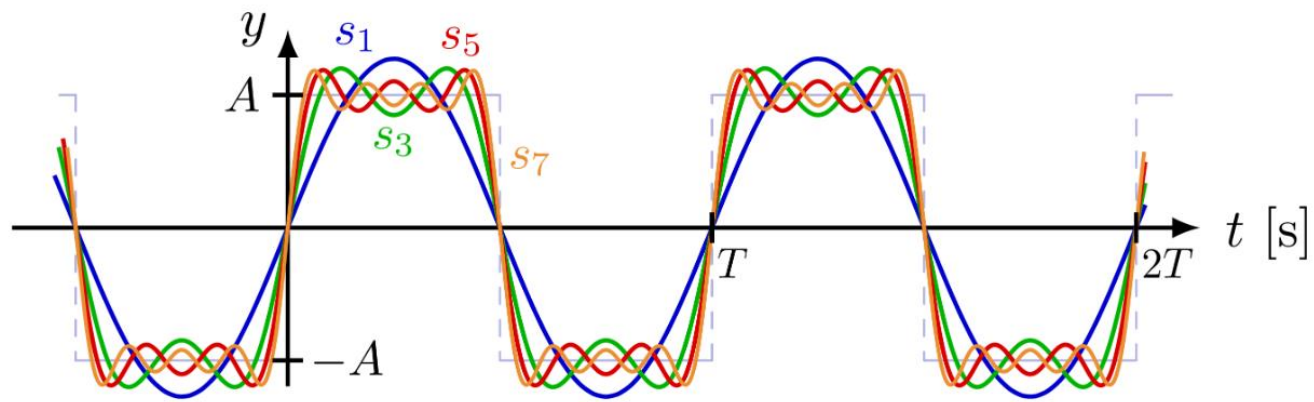
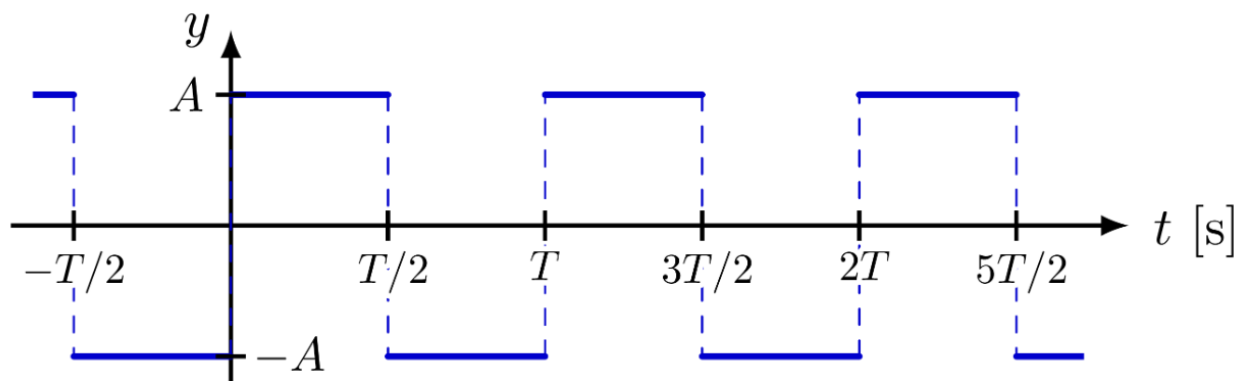
(d) $\cos(4\pi t)$

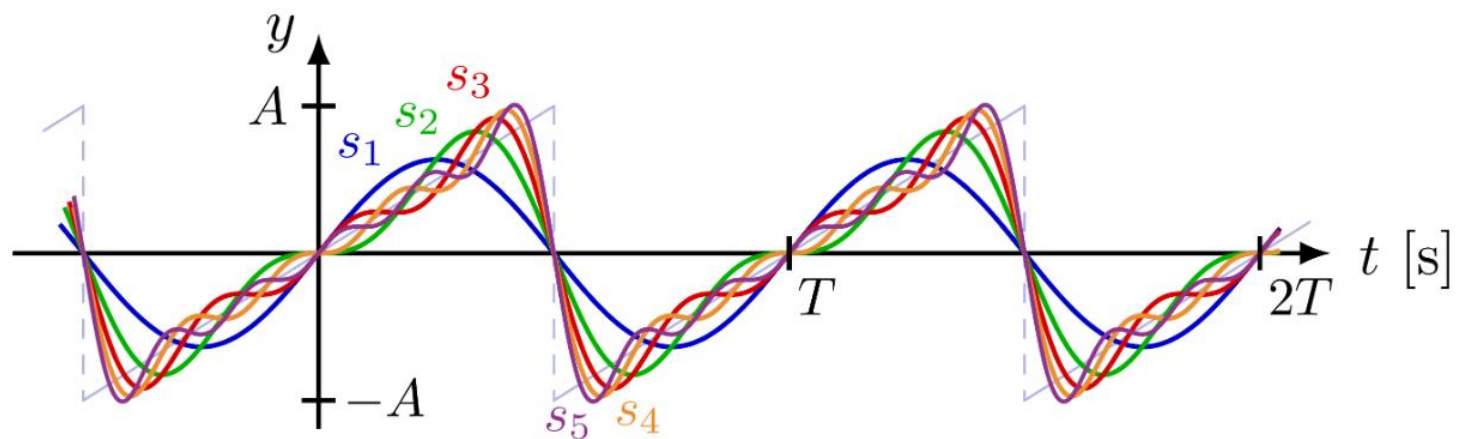
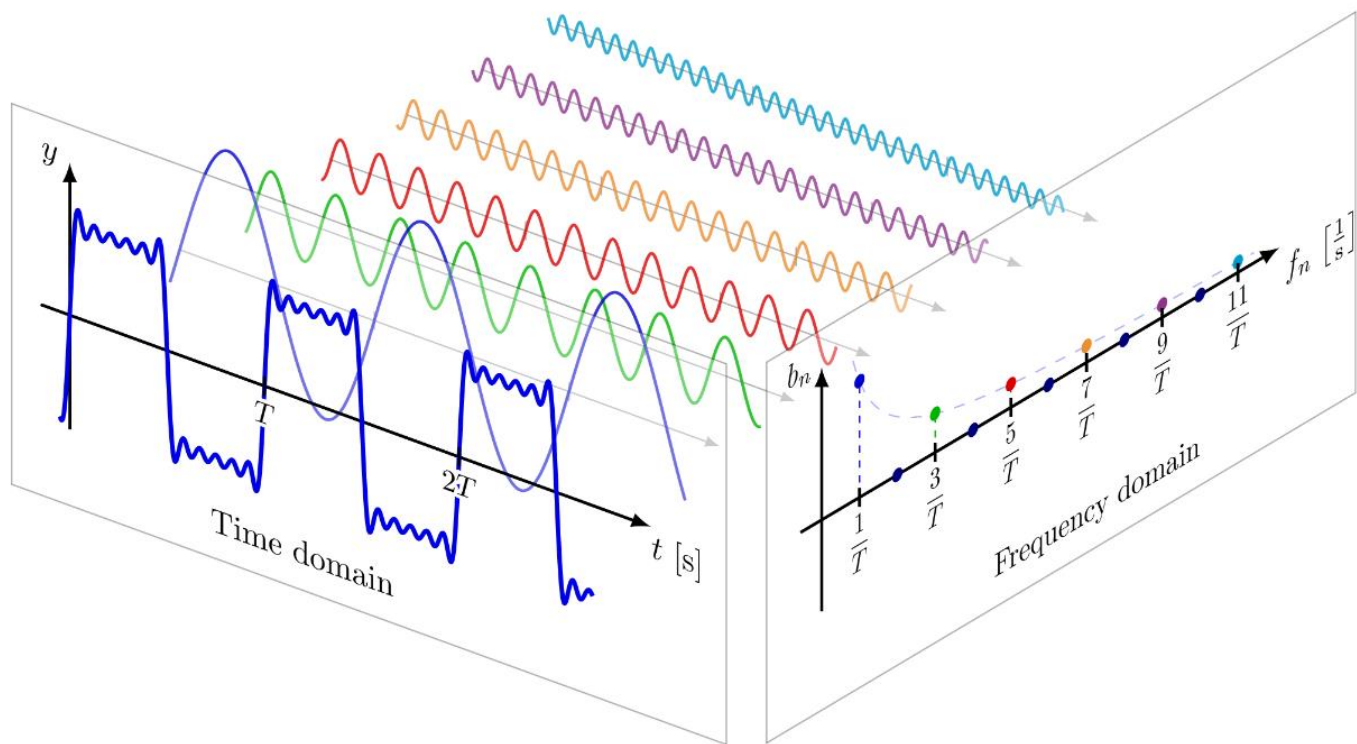


(e) $\sin(4\pi t)$



(f) $\cos(6\pi t)$







Orthogonal Series Expansion

❖ Suppose $\{\phi_n(x)\}$ is an orthogonal set on $[a, b]$. If $f(x)$ is defined on $[a, b]$, we first write as

$$f(x) = c_0\phi_0(x) + c_1\phi_1(x) + \cdots + c_n\phi_n(x) + \cdots? \quad (6)$$

Then $\int_a^b f(x)\phi_m(x) dx$

$$= c_0 \int_a^b \phi_0(x)\phi_m(x) dx + c_1 \int_a^b \phi_1(x)\phi_m(x) dx + \cdots$$

$$+ c_n \int_a^b \phi_n(x)\phi_m(x) dx + \cdots$$

$$= c_0(\phi_0, \phi_m) + c_1(\phi_1, \phi_m) + \cdots + c_n(\phi_n, \phi_m) + \cdots$$



❖ Since $\{\phi_n(x)\}$ is an orthogonal set on $[a, b]$, each term on the right-hand side is zero except $m = n$. In this case we have

$$\int_a^b f(x)\phi_n(x)dx = c_n(\phi_n, \phi_n) = c_n \int_a^b \phi_n^2(x)dx$$

$$c_n = \frac{\int_a^b f(x)\phi_n(x)dx}{\int_a^b \phi_n^2(x)dx}, \quad n = 0, 1, 2, \dots$$



In other words,

$$f(x) = \sum_{n=0}^{\infty} c_n \phi_n(x), \quad (7)$$

$$c_n = \frac{\int_a^b f(x) \phi_n(x) dx}{\|\phi_n(x)\|^2} \quad (8)$$

Then (7) becomes

$$f(x) = \sum_{n=0}^{\infty} \frac{(f, \phi_n)}{\|\phi_n(x)\|^2} \phi_n(x) \quad (9)$$



12.2 Fourier Series

❖ Trigonometric Series

We can show that the set

$$\left\{ 1, \cos \frac{\pi}{p} x, \cos \frac{2\pi}{p} x, \cos \frac{3\pi}{p} x, \dots, \sin \frac{\pi}{p} x, \sin \frac{2\pi}{p} x, \sin \frac{3\pi}{p} x, \dots \right\} \quad (1)$$

is **orthogonal on $[-p, p]$** . Thus a function f defined on $[-p, p]$ can be written as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi}{p} x + b_n \sin \frac{n\pi}{p} x \right) \quad (2)$$



❖ Now we calculate the coefficients.

$$\int_{-p}^p f(x)dx = \frac{a_0}{2} \int_{-p}^p dx + \sum_{n=1}^{\infty} \left(a_n \int_{-p}^p \cos \frac{n\pi}{p} x dx + b_n \int_{-p}^p \sin \frac{n\pi}{p} x dx \right) \quad (3)$$

Since $\cos(n\pi x/p)$ and $\sin(n\pi x/p)$ are orthogonal to 1 on this interval, then (3) becomes

$$\int_{-p}^p f(x)dx = \frac{a_0}{2} \int_{-p}^p dx = \frac{a_0}{2} x \Big|_{-p}^p = pa_0$$

Thus we have

$$a_0 = \frac{1}{p} \int_{-p}^p f(x)dx \quad (4)$$



❖ In addition,

$$\begin{aligned} & \int_{-p}^p f(x) \cos \frac{m\pi}{p} x dx \\ &= \frac{a_0}{2} \int_{-p}^p \cos \frac{m\pi}{p} x dx + \\ & \sum_{n=1}^{\infty} \left(a_n \int_{-p}^p \cos \frac{m\pi}{p} x \cos \frac{n\pi}{p} x dx + b_n \int_{-p}^p \cos \frac{m\pi}{p} x \sin \frac{n\pi}{p} x dx \right) \end{aligned} \quad (5)$$

by orthogonality we have

$$\int_{-p}^p \cos \frac{m\pi}{p} x dx = 0, m > 0$$

$$\int_{-p}^p \cos \frac{m\pi}{p} x \sin \frac{n\pi}{p} x dx = 0$$



❖ and

$$\int_{-p}^p \cos \frac{m\pi}{p} x \cos \frac{n\pi}{p} x dx = \begin{cases} 0, & m \neq n \\ p, & m = n \end{cases}$$

Thus (5) reduces to

$$\int_{-p}^p f(x) \cos \frac{n\pi}{p} x dx = a_n p$$

and so

$$a_n = \frac{1}{p} \int_{-p}^p f(x) \cos \frac{n\pi}{p} x dx \quad (6)$$



❖ Finally, if we multiply (2) by $\sin(m\pi x/p)$ and use

$$\int_{-p}^p \sin \frac{m\pi}{p} x dx = 0, \quad m > 0$$

$$\int_{-p}^p \sin \frac{m\pi}{p} x \sin \frac{n\pi}{p} x dx = 0$$

and

$$\int_{-p}^p \sin \frac{m\pi}{p} x \sin \frac{n\pi}{p} x dx = \begin{cases} 0, & m \neq n \\ p, & m = n \end{cases}$$

we find that

$$\int_{-p}^p \sin \frac{m\pi}{p} x \sin \frac{n\pi}{p} x dx = \begin{cases} 0, & m \neq n \\ p, & m = n \end{cases} \quad (7)$$

Fourier Series

The **Fourier series** of a function f defined on the interval $(-p, p)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi}{p} x + b_n \sin \frac{n\pi}{p} x \right) \quad (8)$$

where

$$a_0 = \frac{1}{p} \int_{-p}^p f(x) dx \quad (9)$$

$$a_n = \frac{1}{p} \int_{-p}^p f(x) \cos \frac{n\pi}{p} x dx \quad (10)$$

$$b_n = \frac{1}{p} \int_{-p}^p f(x) \sin \frac{n\pi}{p} x dx \quad (11)$$



Example 1

Expand $f(x) = \begin{cases} 0, & -\pi < x < 0 \\ \pi - x, & 0 \leq x < \pi \end{cases}$ (12)
in a Fourier series.

Solution

The graph of f is shown in Fig 12.1 with $p = \pi$.

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \left[\int_{-\pi}^0 0 dx + \int_0^{\pi} (\pi - x) dx \right] \\ &= \frac{1}{\pi} \left[\pi x - \frac{x^2}{2} \right]_0^{\pi} = \frac{\pi}{2} \end{aligned}$$



Fig 12.1

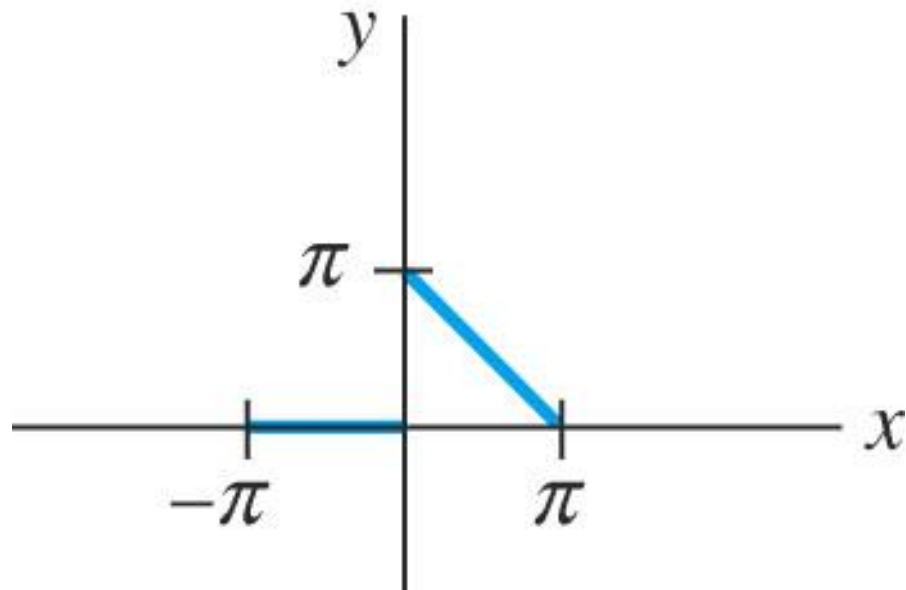
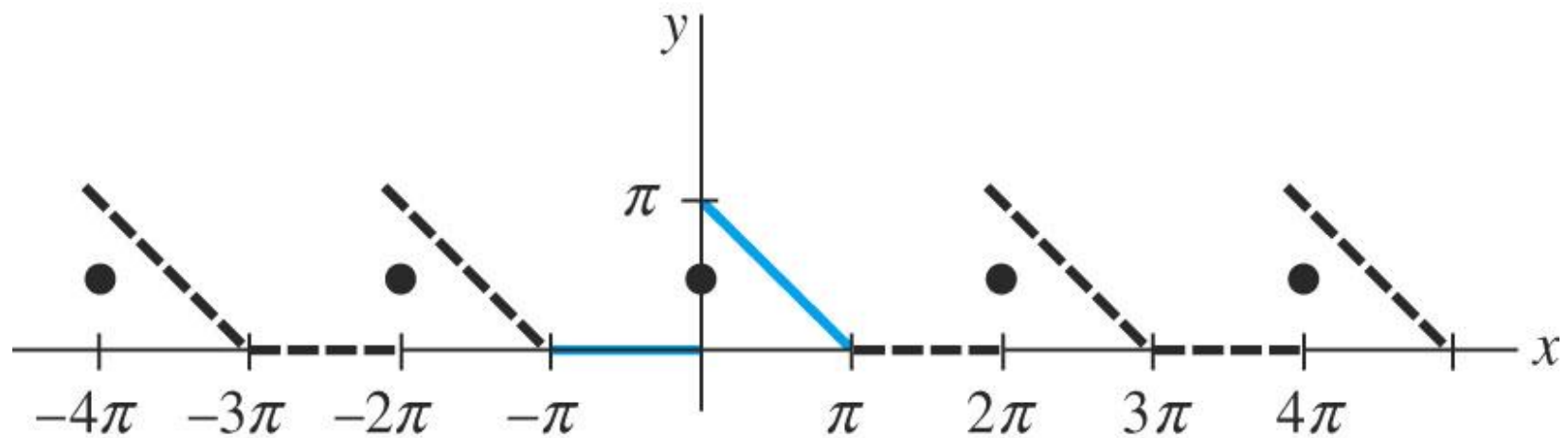




Fig 12.2





Example 1 (2)

$$\begin{aligned}a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx \\&= \frac{1}{\pi} \left[\int_{-\pi}^0 0 dx + \int_0^{\pi} (\pi - x) \cos nx \, dx \right] \\&= \frac{1}{\pi} \left[(\pi - x) \frac{\sin nx}{n} \Big|_0^{\pi} + \frac{1}{n} \int_0^{\pi} \sin nx \, dx \right] \\&= -\frac{1}{n\pi} \frac{\cos nx}{n} \Big|_0^{\pi} \quad \leftarrow \cos n\pi = (-1)^n \\&= \frac{-\cos n\pi + 1}{n^2 \pi} = \frac{1 - (-1)^n}{n^2 \pi}\end{aligned}$$



Example 1 (3)

From (11) we have

$$b_n = \frac{1}{\pi} \int_0^\pi (\pi - x) \sin nx dx = \frac{1}{n}$$

Therefore

$$f(x) = \frac{\pi}{4} + \sum_{n=1}^{\infty} \left\{ \frac{1 - (-1)^n}{n^2 \pi} \cos nx + \frac{1}{n} \sin nx \right\} \quad (13)$$



Sequence of Partial Sums

❖ Sequence of Partial Sums

Referring to (13), we write the partial sums as

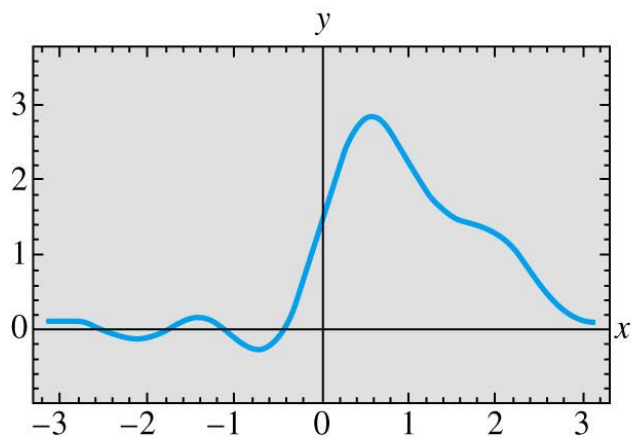
$$S_1 = \frac{\pi}{4}, \quad S_2 = \frac{\pi}{4} + \frac{2}{\pi} \cos x + \sin x,$$

$$S_3 = \frac{\pi}{4} + \frac{2}{\pi} \cos x + \sin x + \frac{1}{2} \sin 2x$$

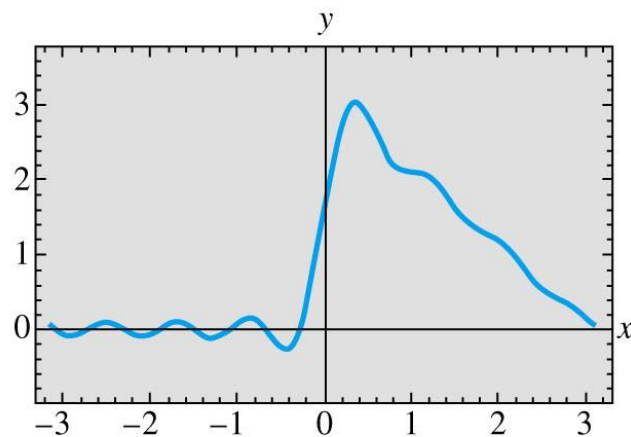
See Fig 12.3.



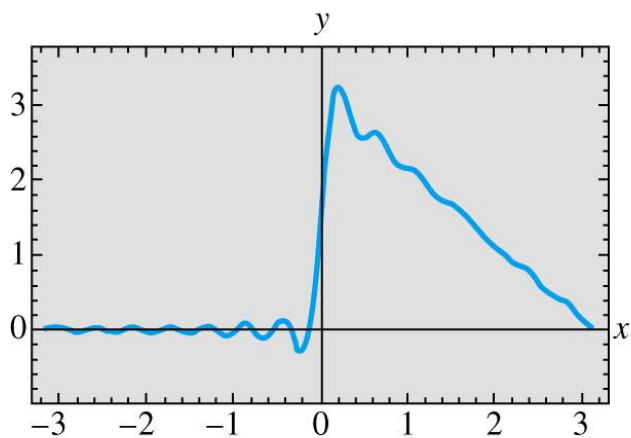
Fig 12.3



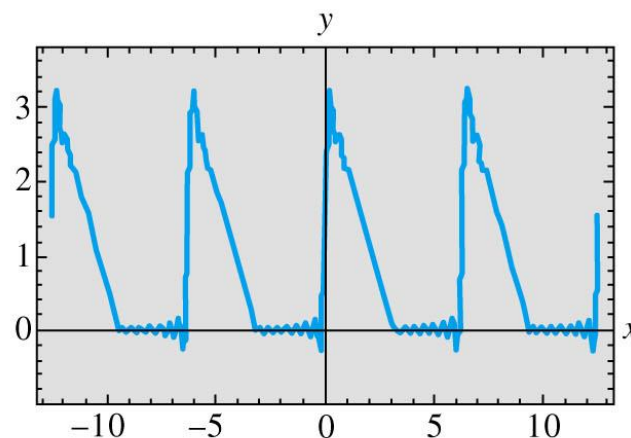
(a) $S_5(x)$ on $(-\pi, \pi)$



(b) $S_8(x)$ on $(-\pi, \pi)$



(c) $S_{15}(x)$ on $(-\pi, \pi)$



(d) $S_{15}(x)$ on $(-4\pi, 4\pi)$



12.3 Fourier Cosine and Sine Series

❖ Even and Odd Functions

- even if $f(-x) = f(x)$
- odd if $f(-x) = -f(x)$



Fig 12.4 Even function

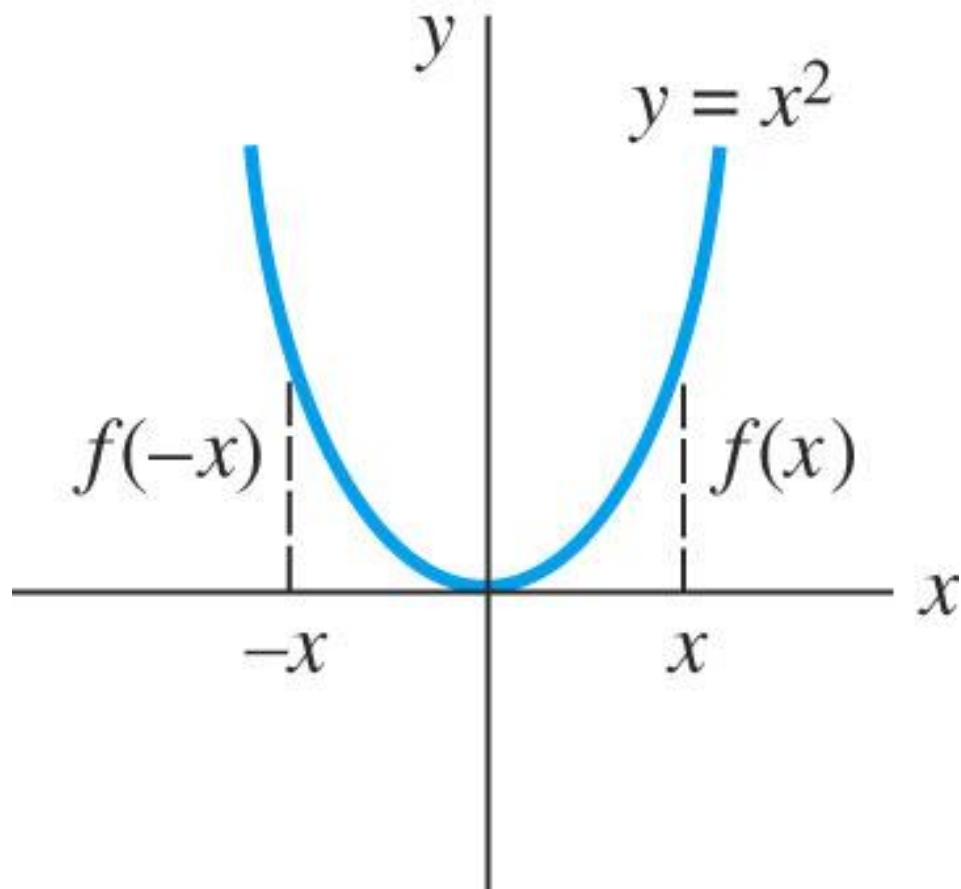
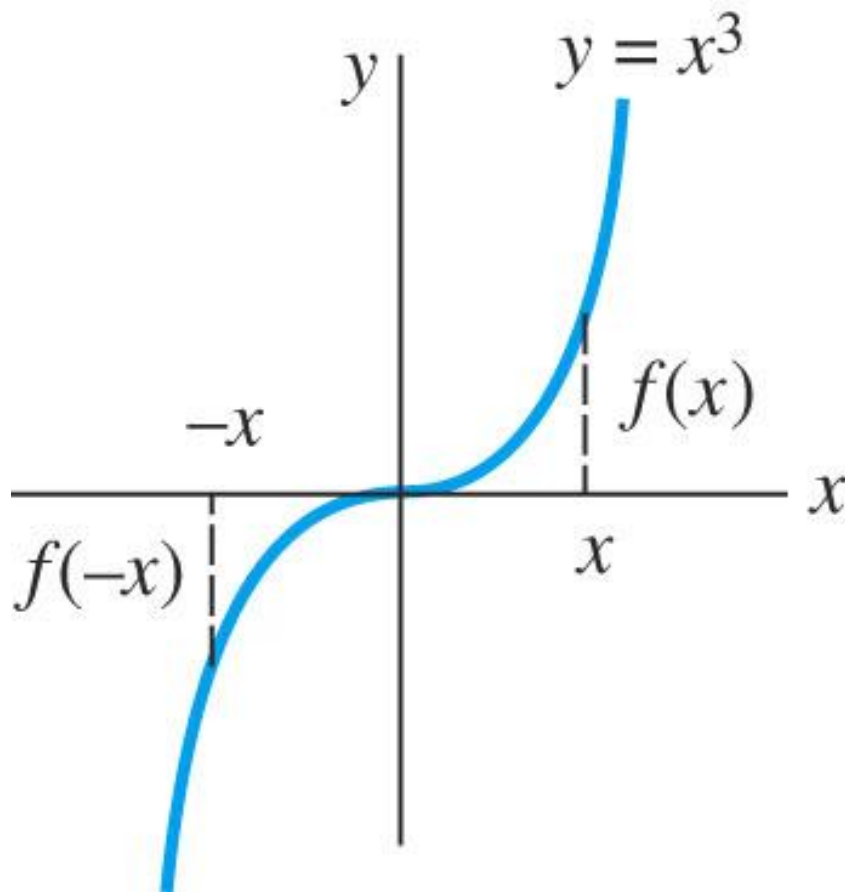




Fig 12.5 Odd function





THEOREM 12.2

Properties of Even/Odd Functions

- (a) The product of two even functions is even.
- (b) The product of two odd functions is even.
- (c) The product of an even function and an odd function is odd.
- (d) The sum (difference) of two even functions is even.
- (e) The sum (difference) of two odd functions is odd.
- (f) If f is even then $\int_{-a}^a f(x)dx = 2\int_0^a f(x)dx$
- (g) If f is odd then $\int_{-a}^a f(x)dx = 0$



Cosine and Sine Series

❖ If f is even on $(-p, p)$ then

$$a_0 = \frac{1}{p} \int_{-p}^p f(x) dx = \frac{2}{p} \int_0^p f(x) dx$$

$$a_n = \frac{1}{p} \int_{-p}^p f(x) \cos \frac{n\pi}{p} x dx = \frac{2}{p} \int_0^p f(x) \cos \frac{n\pi}{p} x dx$$

$$b_n = \frac{1}{p} \int_{-p}^p f(x) \sin \frac{n\pi}{p} x dx = 0$$

Similarly, if f is odd on $(-p, p)$ then

$$a_n = 0, n = 0, 1, 2, \dots \quad b_n = \frac{2}{p} \int_0^p f(x) \sin \frac{n\pi}{p} x dx$$



DEFINITION 12.6

Fourier Cosine and Sine Series

(i) The Fourier series of an even function f on the interval $(-p, p)$ is the **cosine series**

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi}{p} x \quad (1)$$

where

$$a_0 = \frac{2}{p} \int_0^p f(x) dx \quad (2)$$

$$a_n = \frac{2}{p} \int_0^p f(x) \cos \frac{n\pi}{p} x dx \quad (3)$$



(continued)

DEFINITION 12.6

Fourier Cosine and Sine Series

(ii) The Fourier series of an odd function f on the interval $(-p, p)$ is the sine series

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi}{p} x \quad (4)$$

where

$$b_n = \frac{2}{p} \int_0^p f(x) \sin \frac{n\pi}{p} x \, dx \quad (5)$$



Example 1

Expand $f(x) = x$, $-2 < x < 2$ in a Fourier series.

Solution

Inspection of Fig 12.6, we find it is an odd function on $(-2, 2)$ and $p = 2$.

$$b_n = \int_0^2 x \sin \frac{n\pi}{2} x \, dx = \frac{4(-1)^{n+1}}{n\pi}$$

Thus

$$f(x) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin \frac{n\pi}{2} x \quad (6)$$

Fig 12.7 is the periodic extension of the function in Example 1.



Fig 12.6

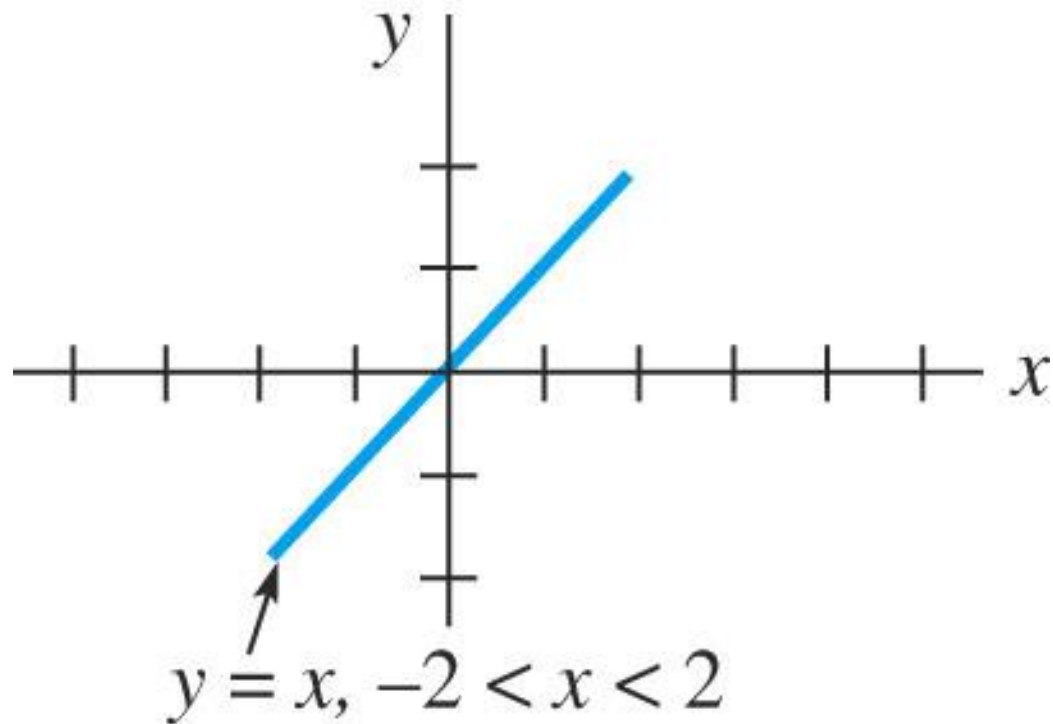
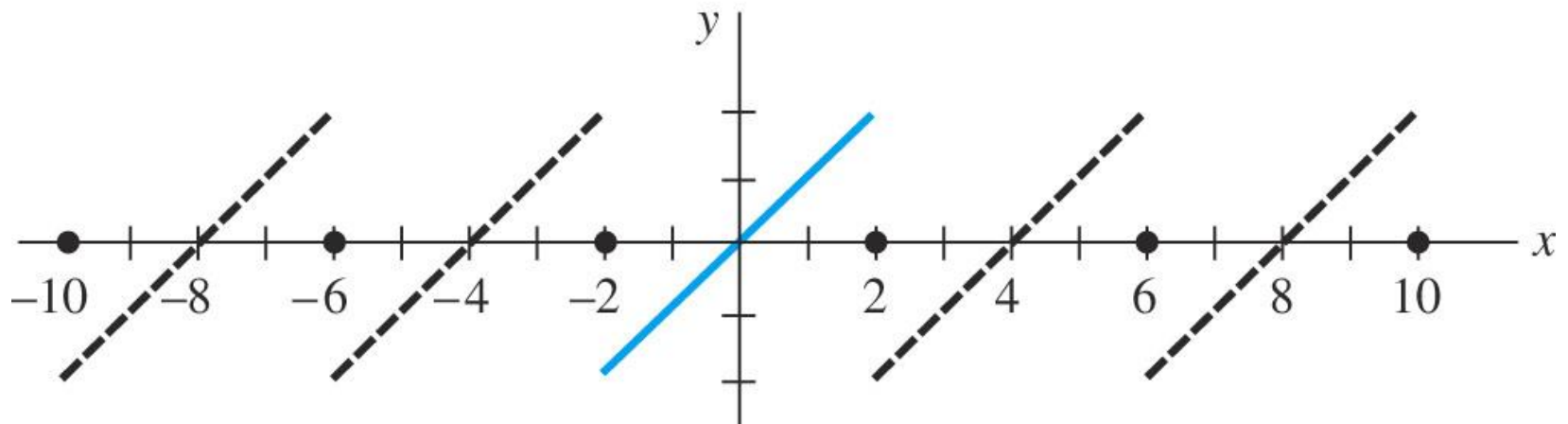




Fig 12.7





Example 2

❖ The function

$$f(x) = \begin{cases} -1, & -\pi < x < 0 \\ 1, & 0 \leq x < \pi \end{cases}$$

shown in Fig 12.8 is odd on $(-\pi, \pi)$ with $p = \pi$.
From (5),

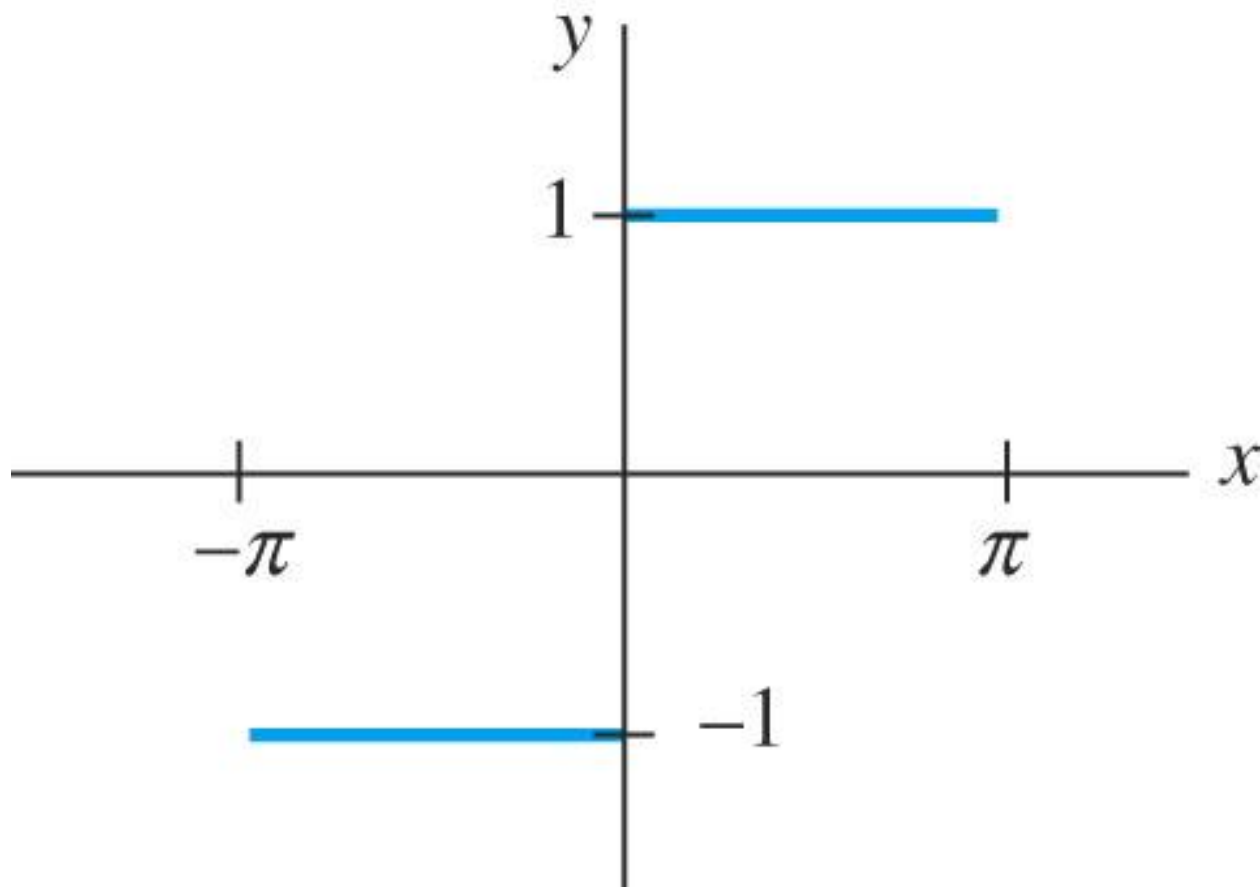
$$b_n = \frac{2}{\pi} \int_0^{\pi} (1) \sin nx \, dx = \frac{2}{\pi} \frac{1 - (-1)^n}{n}$$

and so

$$f(x) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n} \sin nx \quad (7)$$



Fig 12.8



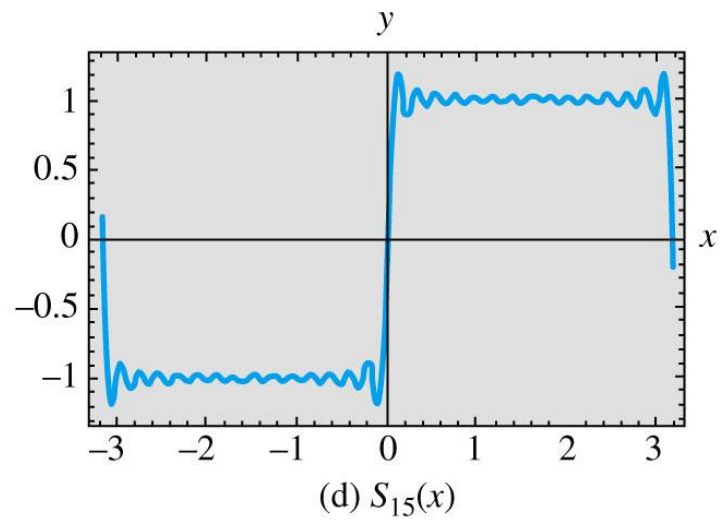
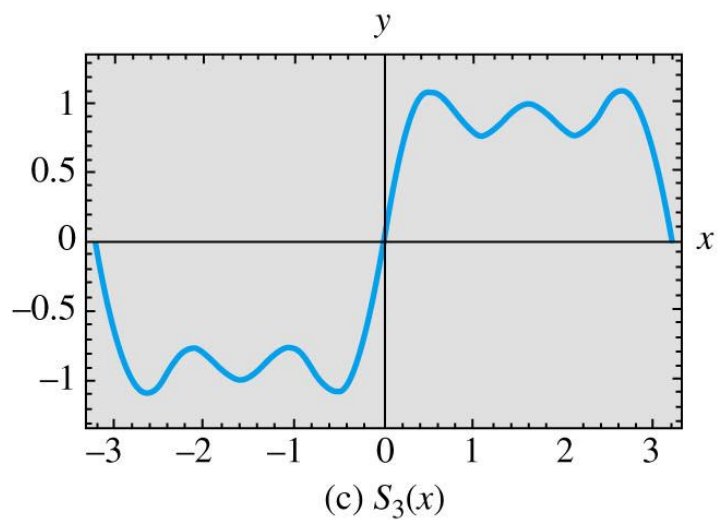
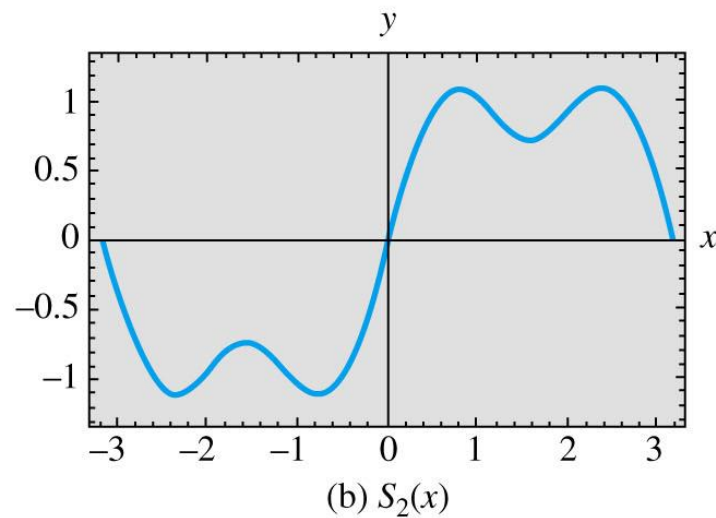
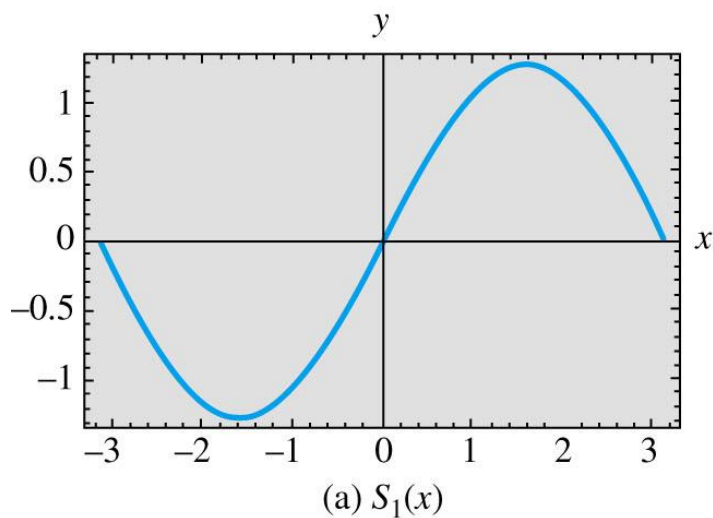


Gibbs Phenomenon

- ❖ Fig 12.9 shows the partial sums of (7).
We can see there are pronounced spikes near the discontinuities. This overshooting of S_N does not smooth out but remains fairly constant even when N is large. This is so-called **Gibbs phenomenon**.



Fig 12.9





12.4 Complex Fourier Series

❖ Euler's formula

$$e^{ix} = \cos x + i \sin x$$

$$e^{-ix} = \cos x - i \sin x \quad (1)$$



Complex Fourier Series

❖ From (1), we have

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}, \quad \sin x = \frac{e^{ix} - e^{-ix}}{2i} \quad (2)$$

Using (2) to replace $\cos(n\pi x/p)$ and $\sin(n\pi x/p)$, then

$$\begin{aligned} & \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \frac{e^{in\pi x/p} + e^{-in\pi x/p}}{2} + b_n \frac{e^{in\pi x/p} - e^{-in\pi x/p}}{2i} \right] \\ &= \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[\frac{1}{2} (a_n - ib_n) e^{in\pi x/p} + \frac{1}{2} (a_n + ib_n) e^{-in\pi x/p} \right] \\ &= c_0 + \sum_{n=1}^{\infty} c_n e^{in\pi x/p} + \sum_{n=1}^{\infty} c_{-n} e^{-in\pi x/p} \end{aligned} \quad (3)$$



where $c_0 = a_0/2$, $c_n = (a_n - ib_n)/2$, $c_{-n} = (a_n + ib_n)/2$.
When the function f is real, c_n and c_{-n} are complex conjugates.

We have

$$c_0 = \frac{1}{2} \cdot \frac{1}{p} \int_{-p}^p f(x) dx \quad (4)$$



$$\begin{aligned} c_n &= \frac{1}{2}(a_n - ib_n) \\ &= \frac{1}{2} \left(\frac{1}{p} \int_{-p}^p f(x) \cos \frac{n\pi}{p} x \, dx - i \frac{1}{p} \int_{-p}^p f(x) \sin \frac{n\pi}{p} x \, dx \right) \\ &= \frac{1}{2p} \int_{-p}^p f(x) \left[\cos \frac{n\pi}{p} x - i \sin \frac{n\pi}{p} x \right] dx \\ &= \frac{1}{2} \int_{-p}^p f(x) e^{-in\pi x/p} dx \end{aligned} \tag{5}$$



$$\begin{aligned} c_{-n} &= \frac{1}{2}(a_n + ib_n) \\ &= \frac{1}{2} \left(\frac{1}{p} \int_{-p}^p f(x) \cos \frac{n\pi}{p} x \, dx + i \frac{1}{p} \int_{-p}^p f(x) \sin \frac{n\pi}{p} x \, dx \right) \\ &= \frac{1}{2p} \int_{-p}^p f(x) \left[\cos \frac{n\pi}{p} x + i \sin \frac{n\pi}{p} x \right] dx \\ &= \frac{1}{2} \int_{-p}^p f(x) e^{in\pi x/p} dx \end{aligned} \tag{6}$$



DEFINITION 12.7

Complex Fourier Series

The ***Complex Fourier Series*** of function f defined on an interval $(-p, p)$ is given by

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/p} \quad (7)$$

where

$$c_n = \frac{1}{2p} \int_{-p}^p f(x) e^{-in\pi x/p} dx, \quad n = 0, \pm 1, \pm 2, \dots \quad (8)$$



Example 1

Expand $f(x) = e^{-x}$, $-\pi < x < \pi$, in a complex Fourier series.

Solution

with $p = \pi$, (8) gives

$$\begin{aligned} c_n &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-x} e^{-inx} dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-(in+1)x} dx \\ &= \frac{1}{2\pi(in+1)} [e^{-(in+1)\pi} - e^{(in+1)\pi}] \end{aligned}$$



Example 1 (2)

Using Euler's formula

$$e^{-(in+1)\pi} = e^{-\pi} (\cos n\pi - i \sin n\pi) = (-1)^n e^{-\pi}$$

$$e^{(in+1)\pi} = e^{\pi} (\cos n\pi + i \sin n\pi) = (-1)^n e^{\pi}$$

Hence

$$c_n = (-1)^n \frac{(e^{\pi} - e^{-\pi})}{2(in+1)\pi} = (-1)^n \frac{\sinh \pi}{\pi} \frac{1-in}{n^2+1} \quad (9)$$



Example 1 (3)

The complex Fourier series is then

$$f(x) = \frac{\sinh \pi}{\pi} \sum_{n=-\infty}^{\infty} (-1)^n \frac{1-in}{n^2+1} e^{inx} \quad (10)$$

The series (10) converges to the 2π -periodic extension of f .



Fundamental Frequency

- ❖ The fundamental period is $T = 2p$ and then $p = T/2$.
The Fourier series becomes

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\omega x + b_n \sin n\omega x) \quad (11)$$

$$\sum_{n=-\infty}^{\infty} c_n e^{in\omega x}$$

where $\omega = 2\pi/T$ is called the ***fundamental angular frequency***.



Frequency Spectrum

- ❖ If f is periodic and has fundamental period T , the plot of the points $(n\omega, |c_n|)$ is called the ***frequency spectrum*** of f .



Example 2

❖ In Example 1, $\omega = 1$, so that $n\omega$ takes on the values $0, \pm 1, \pm 2, \dots$

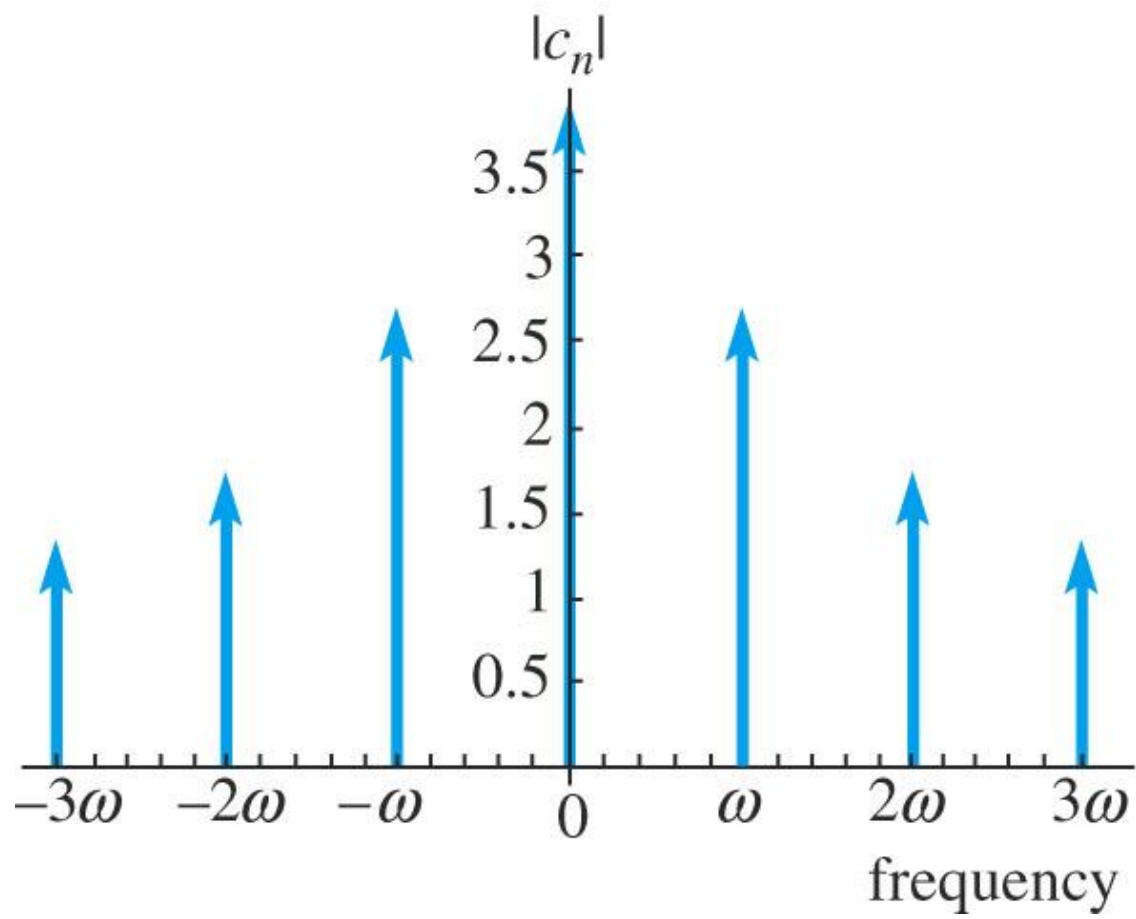
Using $|\alpha + i\beta| = \sqrt{\alpha^2 + \beta^2}$, we see from (9) that

$$|c_n| = \frac{\sinh \pi}{\pi} \frac{1}{\sqrt{n^2 + 1}}$$

See Fig 12.17.



Fig 12.17

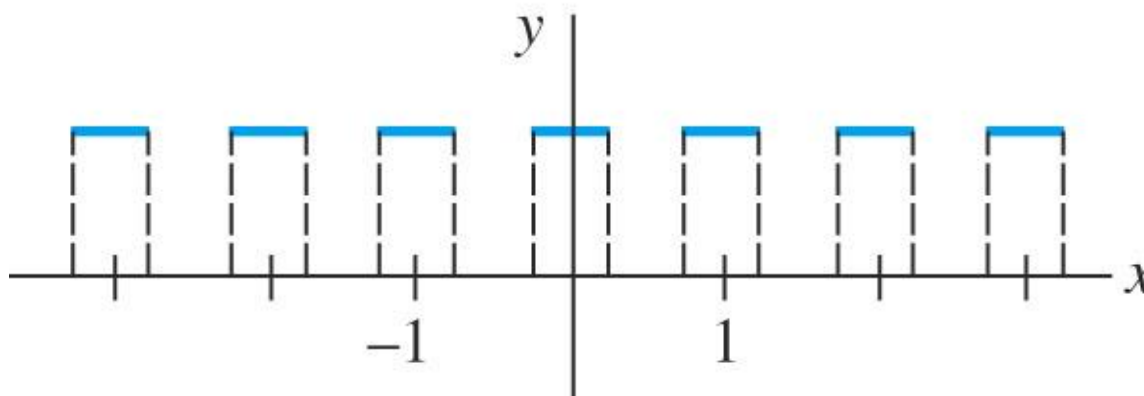




Example 3

- ❖ Find the spectrum of the wave shown in Fig12.18.
The wave is the periodic extension of the function f :

$$f(x) = \begin{cases} 0, & -\frac{1}{2} < x < -\frac{1}{4} \\ 1, & -\frac{1}{4} < x < \frac{1}{4} \\ 0, & \frac{1}{4} < x < \frac{1}{2} \end{cases}$$





Example 3 (2)

Solution

Here $T = 1 = 2p$ so $p = 1/2$. Since f is 0 on $(-1/2, -1/4)$ and $(1/4, 1/2)$, (8) becomes

$$\begin{aligned} c_n &= \int_{-1/2}^{1/2} f(x) e^{2in\pi x} dx = \int_{-1/4}^{1/4} (1) e^{2in\pi x} dx \\ &= \left. \frac{e^{2in\pi x}}{2in\pi} \right|_{-1/4}^{1/4} = \frac{1}{n\pi} \frac{e^{in\pi/2} - e^{-in\pi/2}}{2i} \\ c_n &= \frac{1}{n\pi} \sin \frac{n\pi}{2} \end{aligned}$$



Example 3 (3)

It is easy to check that

$$c_0 = \int_{-1/4}^{1/4} dx = \frac{1}{2}$$

Fig 12.19 shows the frequency spectrum of f .



Fig 12.19

